GOLDEN GATE
UNIVERSITY

Database Systems

Lecture 4 – Data Representation, ER Modeling, Schema Design & Data Visualization

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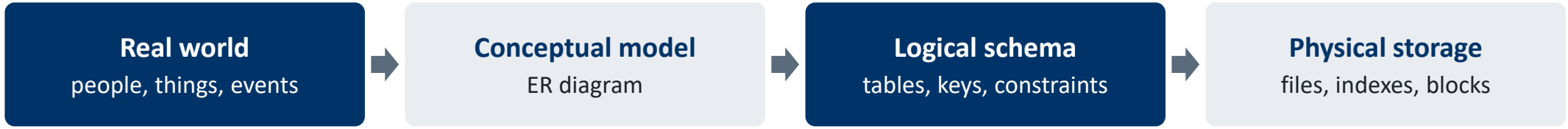
Learning outcomes and agenda

Time	Block
0-9	Data representation and design process
9-30	ER building blocks (case study)
30-38	Full ER diagram, design issues
38-48	ER to schema, schema quality
48-57	Data visualization
57-60	Summary and review

By the end you can:

- Describe data at three levels of abstraction
- Draw an ER diagram from requirements
- Map ER diagrams to tables
- Detect redundancy in a schema
- Pick an honest chart

Data representation basics



Form of data	Description	Example	Typical storage
Structured	Fixed columns, same shape for every record	Student(ID, name, credits)	Relational tables
Semi-structured	Self-describing, flexible attributes	JSON or XML order document	Document stores, JSON columns
Unstructured	No predefined fields	Lecture video, free text, images	Files, blobs, search indexes

Representation = choosing which facts to keep and in what shape.

The database design process

Requirements
talk to users

→

Conceptual design
ER diagram

→

Logical design
relational schema

→

Physical design
indexes, storage

Phase	Question it answers	Output
Requirements	What must the system store and answer?	Written specification
Conceptual	Which entities and relationships exist?	ER diagram
Logical	Which tables, keys and constraints?	Relational schema
Physical	How is it stored for speed?	Index and file choices

Mistakes found early cost minutes; mistakes found after launch cost months.

Running case study: a university

Requirements (plain language)

1. A university has departments with a name, building and budget.
2. Each instructor works in one department and has an ID, name and salary.
3. Each student has an ID, name and total credits, and one advisor (an instructor).
4. A department offers courses (course id, title, credits).
5. A course runs as sections in a semester and year.
6. Students take sections and receive a grade.
7. Instructors teach sections.

Nouns

department, instructor, student, course, section: candidate entities

Verbs

works in, advises, offers, takes, teaches: candidate relationships

Adjectives and details

name, salary, credits, grade: candidate attributes

Entities and attributes

instructor
ID
name
first_name
last_name
salary
{phone_number}
date_of_birth
age()

Attribute type	Meaning	Example
Simple	Cannot be divided	salary
Composite	Has component parts	name = first + last
Multivalued	Many values per entity	{phone_number}
Derived	Computed from other data	age() from date_of_birth
Null	Value unknown or absent	middle name

Underlined = primary key. { } = multivalued. ( ) = derived.

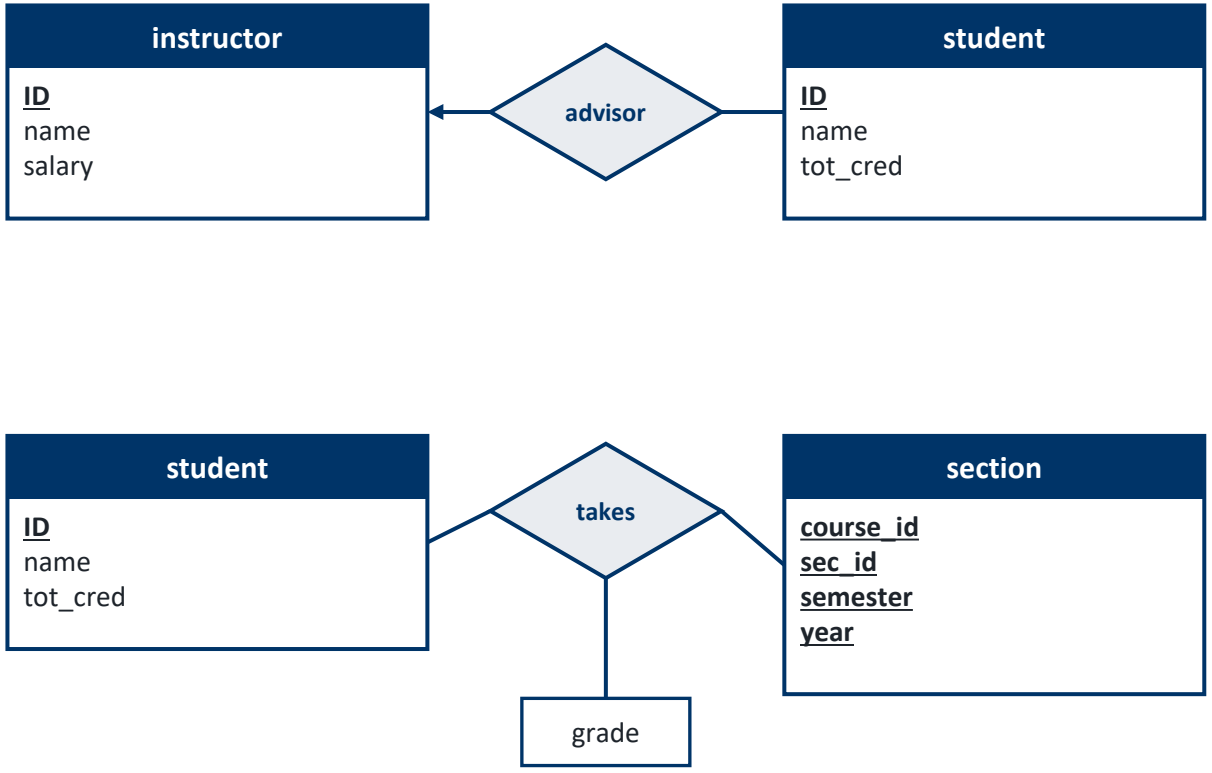
Keys: telling entities apart

Term	Meaning	In instructor
Superkey	Set of attributes that identifies a row uniquely	{ID}, {ID, name}
Candidate key	A minimal superkey	{ID}
Primary key	The candidate key chosen by the designer	ID
Foreign key	Attribute that refers to the primary key of another table	dept_name

Sample relation: instructor

ID	name	salary	dept_name
10101	Srinivasan	65000	Comp. Sci.
12121	Wu	90000	Finance
22222	Einstein	95000	Physics
15151	Mozart	40000	Music

Relationship sets



Key ideas

- A relationship set associates entities from two or more entity sets
- Degree: binary (2) is most common; ternary (3) is rare
- A relationship can carry its own attribute, such as grade
- Role names help when an entity set appears twice

Mapping cardinalities

One-to-one



Each department has one head; each instructor heads at most one

One-to-many



One instructor, many students; a student has one advisor

Many-to-one



Many instructors in one department

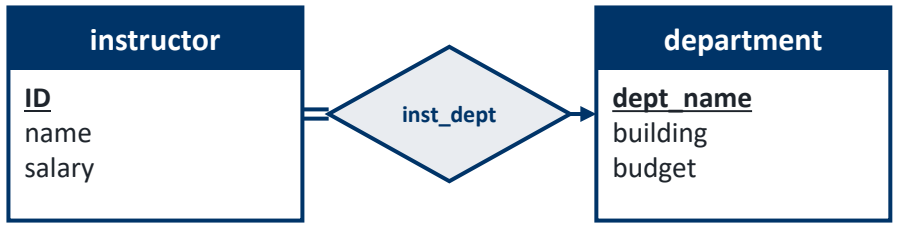
Many-to-many



A student takes many sections; a section has many students

Arrow (→) points to the "one" side. Plain line = "many" side.

Participation and weak entity sets





Total participation

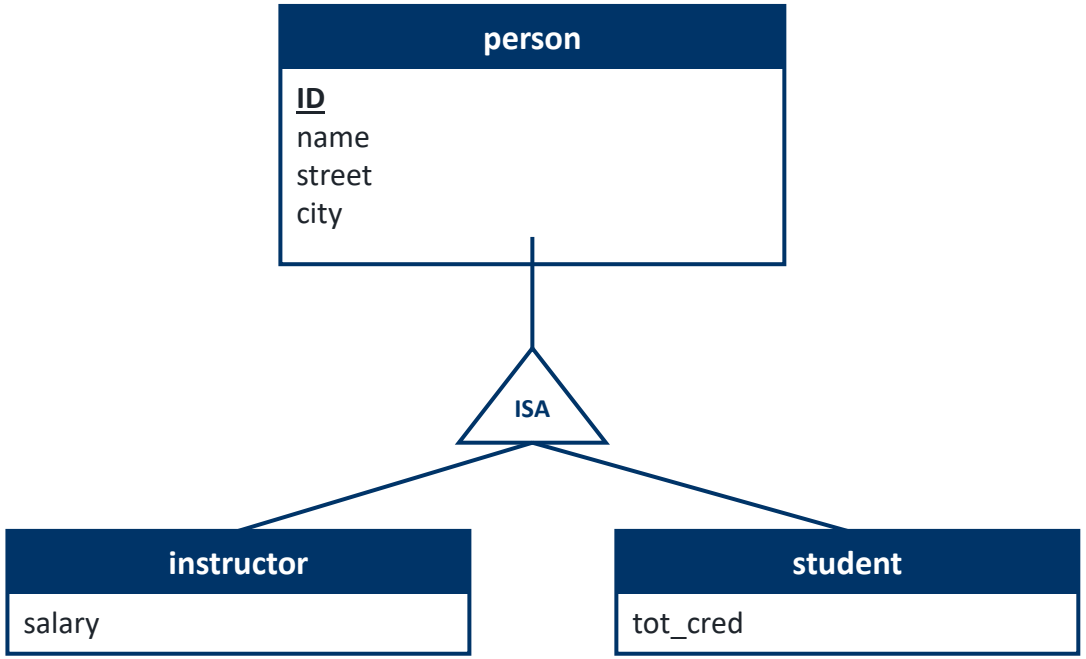
- Every instructor must belong to a department (double line).
- Partial participation: a department may exist with no instructors (single line).

Weak entity set

- Cannot be identified by its own attributes alone.
- Depends on an owner (course) through an identifying relationship (double diamond).
- Key = owner key + discriminator.

Section 1 of which course? sec_id alone is not enough.

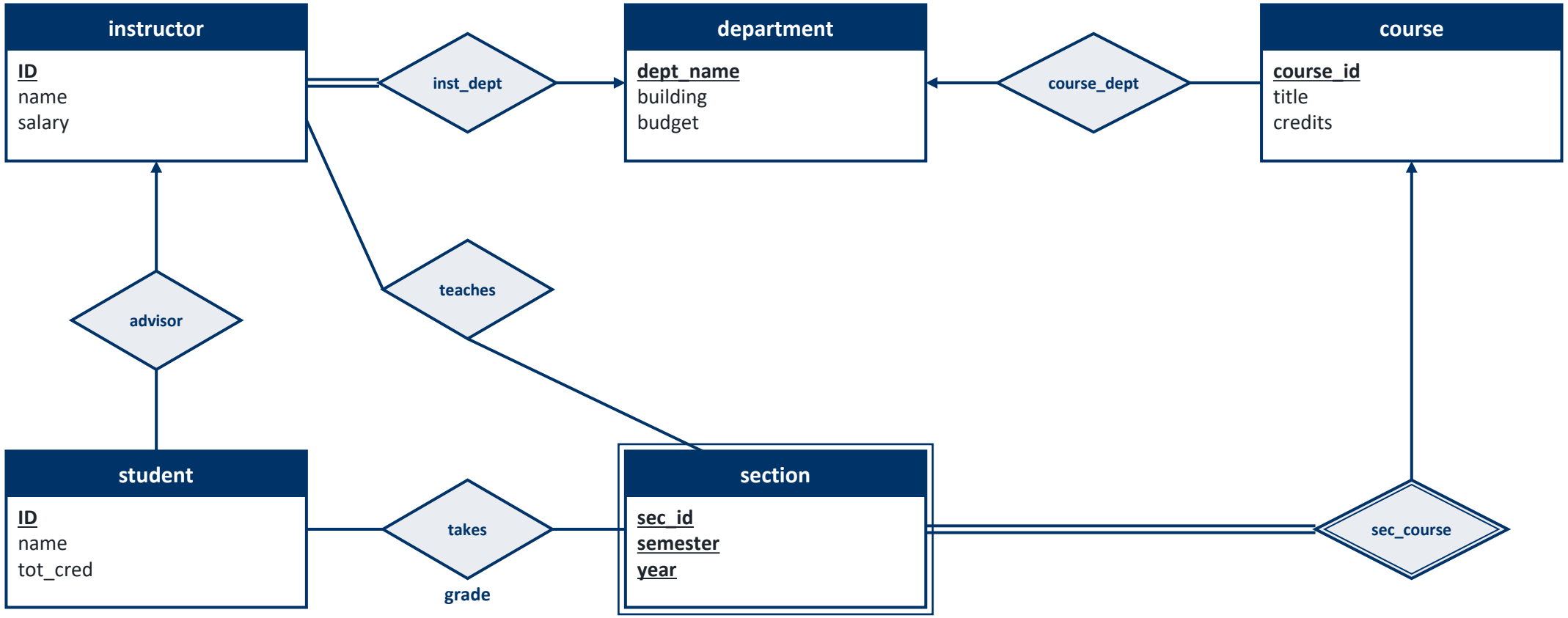
Extended ER: specialization (ISA)



Key ideas

- Specialization (top-down): split an entity set into special groups
- Generalization (bottom-up): merge similar entity sets
- Lower-level sets inherit the attributes of the higher-level set
- Constraints: disjoint or overlapping; total or partial
- Aggregation: treat a relationship as an entity so it can join another relationship

Complete ER diagram: university



Double line = total participation. Arrow = "one" side. Double box / diamond = weak entity.

ER design issues

Decision	Guideline	Example
Attribute or entity set?	Make it an entity set if it has its own attributes or relationships	phone: attribute if only a number; entity if it has type and carrier
Entity set or relationship?	Nouns with identity are entities; actions between them are relationships	"takes" is a relationship, "section" is an entity
Binary or ternary?	Prefer several binary relationships when the facts are independent	instructor-student-project: ternary only if the triple is one fact
Redundant attributes	Do not copy the key of one entity set into another as an attribute	dept_name inside instructor AND inst_dept relationship
Where does an attribute live?	With the entity (or relationship) it truly describes	grade on takes, not on student

From ER diagram to relational schema

ER construct	Rule	University example
Strong entity set	One table; PK = entity key	student(ID, name, tot_cred)
Composite attribute	One column per component	name_first, name_last
Multivalued attribute	Separate table with owner key + value	inst_phone(ID, phone_number)
Derived attribute	Not stored; compute on demand	age from date_of_birth
Weak entity set	Table with owner key + discriminator	section(course_id, sec_id, semester, year)
Many-to-many relationship	Table with both keys (+ attributes)	takes(ID, course_id, ..., grade)
One-to-many / many-to-one	Foreign key on the "many" side	instructor(..., dept_name)
Specialization	Table per entity; child holds parent key	instructor(ID, salary)

The resulting schema (university)

```
department (dept_name*, building, budget)
instructor (ID*, name, salary, dept_name->department)
student    (ID*, name, tot_cred)
course     (course_id*, title, credits,
            dept_name->department)
section    (course_id*->course, sec_id*,
            semester*, year*)
advisor    (s_ID*->student, i_ID->instructor)
takes      (ID*->student, course_id*, sec_id*,
            semester*, year*, grade)
teaches    (ID*->instructor, course_id*, sec_id*,
            semester*, year*)

* = primary key      -> = foreign key
```

Check the logic

- advisor: PK is s_ID because a student has one advisor
- takes and teaches are many-to-many, so each is a table
- sec_course has no table: it is absorbed into section
- dept_name appears as FK only after mapping

Why schema quality matters

ID	name	salary	dept_name	building	budget
22222	Einstein	95000	Physics	Watson	70000
83821	Brandt	92000	Comp. Sci.	**Taylor**	**100000**
45565	Katz	75000	Comp. Sci.	**Taylor**	**100000**

One big table in_dep(ID, name, salary, dept_name, building, budget): Comp. Sci. data is repeated

Redundancy

Building and budget repeat for every instructor of a department

Update anomaly

Change a budget in one row and the others disagree

Insert / delete anomaly

A new department with no instructor cannot be stored; deleting the last instructor erases the department

Fix: instructor(ID, name, salary, dept_name) + department(dept_name, building, budget)

Data visualization overview

Stored data
tables

→

Query (SQL)
GROUP BY, SUM

→

Chart or report
bar, line, map

→

Dashboard
drill down

Reports

Formatted tables with subtotals, built by report generators from queries

Charts

Bar, line, pie and maps encode numbers as length, position and colour

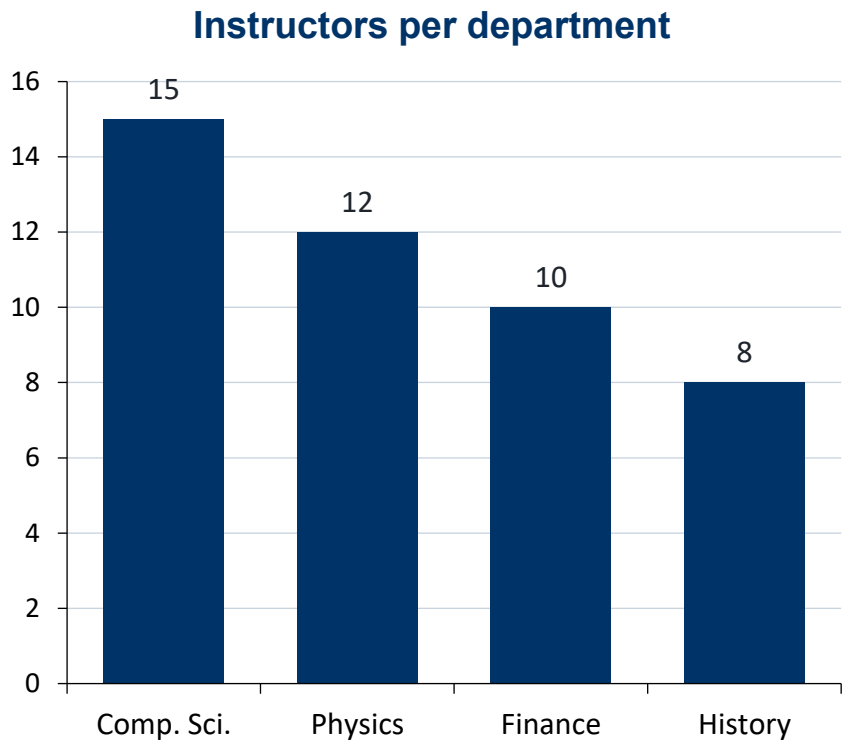
Dashboards

Several charts together; interactive drill-down (Tableau, Plotly, Google Charts)

One graphical screen can carry as much information as many screens of text.

Choosing the right chart

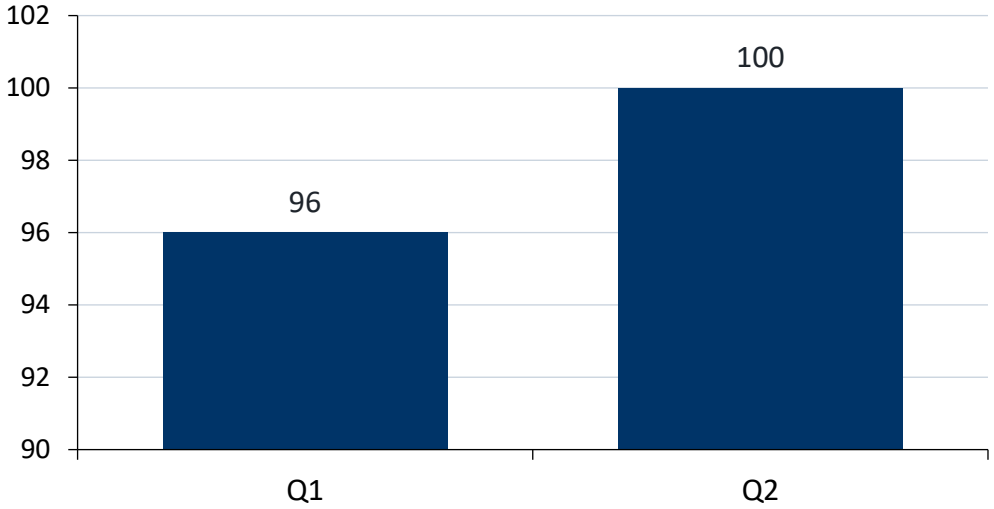
Question	Chart
Compare categories	Bar chart
Change over time	Line chart
Shape of one variable	Histogram
Compare spread between groups	Box plot
Relationship of two numbers	Scatter plot
Few parts of a whole	Stacked bar or pie
Values by place	Map



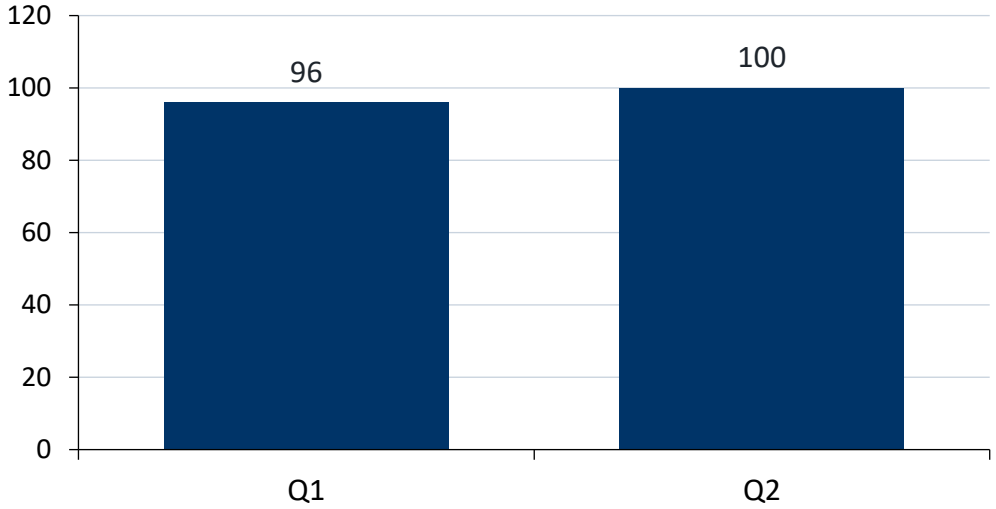
Illustrative data

Honest vs misleading charts

Axis starts at 90: Q2 looks 1.7x taller



Axis starts at 0: 4% growth



Start bar axes at zero

Bar length must match the value

Label axes and units

A reader should not guess

Limit colours and slices

Colour should carry meaning

Avoid 3-D effects

Decoration distorts size

Summary and review questions

Data representation

Real world, conceptual, logical, physical; three forms of data

ER modeling

Entities, attributes, keys, relationships, cardinality, participation, ISA

Schema design

ER to tables by rules; redundancy causes anomalies

Visualization

Query, choose chart, keep it honest

Review questions

- Draw an ER diagram for a library: books, members, loans
- Which cardinality needs its own table?
- What is a weak entity set?
- Name two anomalies of a redundant table

Next lecture: Functional dependencies and normalization (Silberschatz, Chapter 7). Lab: ER design, normalization, visualization, probability in code.